MILITARY SPECIFICATION

FORCEPS, SUTURE TYING (Troutman)

This specification is approved for use within the Defense Personnel Support Center and is available for use by all Departments and Agencies of the Department of Defense.

SCOPE

1.1 Scope. This specification covers a suture tying forceps of the Troutman style.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specification
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of solicitation.

DEFENSE PERSONNEL SUPPORT CENTER (DPSC) DRAWING

23382 - FORCEPS, SUTURE TYING (Troutman)

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.
F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The forceps shall be fabricated from Class 4, martensitic corrosion-resistant steel in accordance with ASTM F899.

3.2 Style, design and dimensions. Forceps shall be of the Troutman design suitable for tying suture during microsurgery. Forceps shall have non-locking jaws which mesh properly when closed. Forceps shall have handle serrations providing a gripping surface. Forceps shall be in accordance with DPSC Drawing No. 23382.

3.3 Hardness. Forceps shall have a Rockwell hardness of C-40 to C-45, when tested as specified in 4.4.1.

3.4 Passivation. Forceps shall be properly passivated.

3.5 Corrosion-resistance. Forceps shall show no corrosion when subjected to the boil test and copper sulfate test specified in 4.4.1 and 4.4.2 respectively.

3.6 Finish. Forceps shall have a mirror or satin finish.

3.7 Identification markings. The forceps shall bear the manufacturer's name or registered trademark and the word "stainless." Instruments of foreign manufacture shall bear the name of the country of manufacture.

3.8 Workmanship. Forceps shall be free from defects which detract from its appearance or impair its serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, or does it commit the Government to acceptance of defective material.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

4.1.3 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.4 Certificate of quality. Certificate of quality supplied by the manufacturer of the corrosion-resistant steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

Sampling.

4.2.1 For examination. Sampling for visual and dimensional examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one forceps.

TABLE I. Sampling criteria for inspection

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purpose shall be one ear forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. Unless otherwise specified, the unit of product shall be one forceps.

TABLE II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Composition Analysis	3.1	ASTM F899	**	*
Hardness test	3.3	ASTM E18	S-1	*
Copper Sulfate test	3.5	GG-I-526	S-1	*
Boil test	3.5	GG-I-526	S-1	*

* Acceptance number is zero.

** Four ounces of each lot of metal used in construction of the forceps shall be selected for chemical analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

4.3 Examination.

4.3.1 Examination for visual and dimensional defects. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Style and design not as specified.
102	Instrument not free from sharp edges, rough areas, burrs, cracks or grind marks.
103	Jaws do not meet properly.
Minor	
201	Identification marking not present or legible.
202	Finish not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Rockwell hardness shall be determined in accordance with ASTM Standard E 18.

4.4.2 Boil test. Forceps shall be subjected to the boil test specified in GG-I-526.

4.4.3 Copper sulfate test. Forceps shall be subjected to the copper sulfate test specified in GG-I-526.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Forceps shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each forceps shall be packaged in accordance with MIL-P-116, Method IC-1.

5.1.1.4 Intermediate package. Twelve unit packages shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for forceps shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for forceps shall conform to the MIL-STD-794 requirements for this level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Forceps packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Forceps packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The forceps of the Troutman design are for use in micro surgery.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

(a) Title, number, and date of specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Applicable National Stock Number.

6.3 National Stock Number Coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6515-01-127-2209	FORCEPS, SUTURE TYING, Troutman

6.4 Subject term (key word) listing.

Forceps
Suture tying forceps
Troutman forceps

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3688

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC-RSTS
2800 South 20th Street
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-42038(DM)	FORCEPS, SUTURE TYING (Troutman)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2026-10-21T20:16Z
Check the source to verify that this is the current version before use.